Text as Pieces

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

AI-1 Week 5: a representation needs agreed encoding rules.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Encode and decode text under an explicit toy vocabulary. Try the recall before opening notes.

Your learning target

I can encode and decode text under an explicit toy **vocabulary**.

I can explain how segmentation changes sequence length.

I can handle unsupported characters without silently deleting information.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

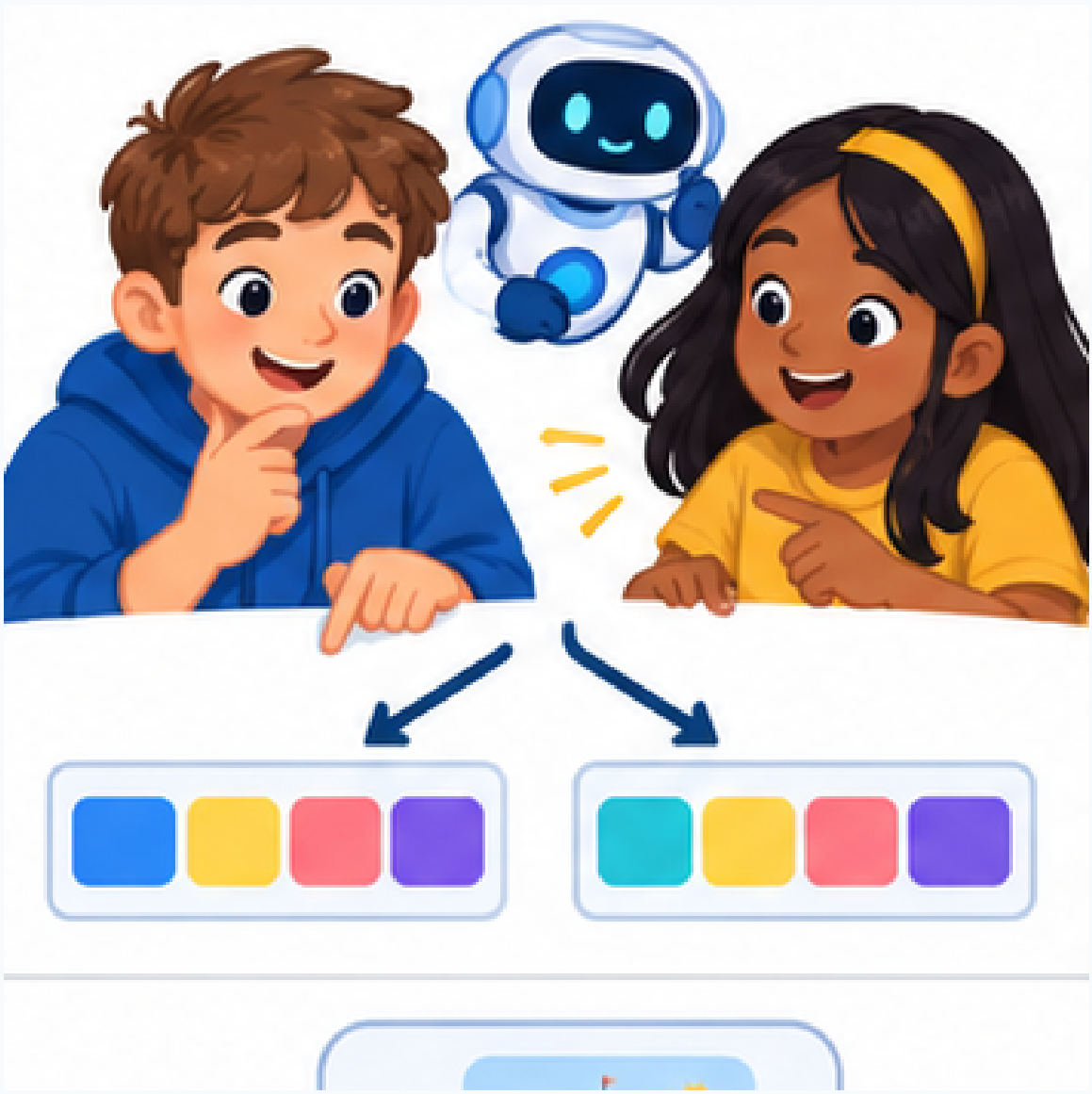
Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 25: Text as Pieces

Language systems **process token**s and **pattern**s rather than whole human meanings.



THINKER MISSION

Tokenize tricky phrases and compare alternate splits.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> What information conditions a next-**token** distribution?

This week's end goal

By the end, I can encode and decode text using a specified **vocabulary** and matching rule.

STUDENT LESSON · 2 OF 18

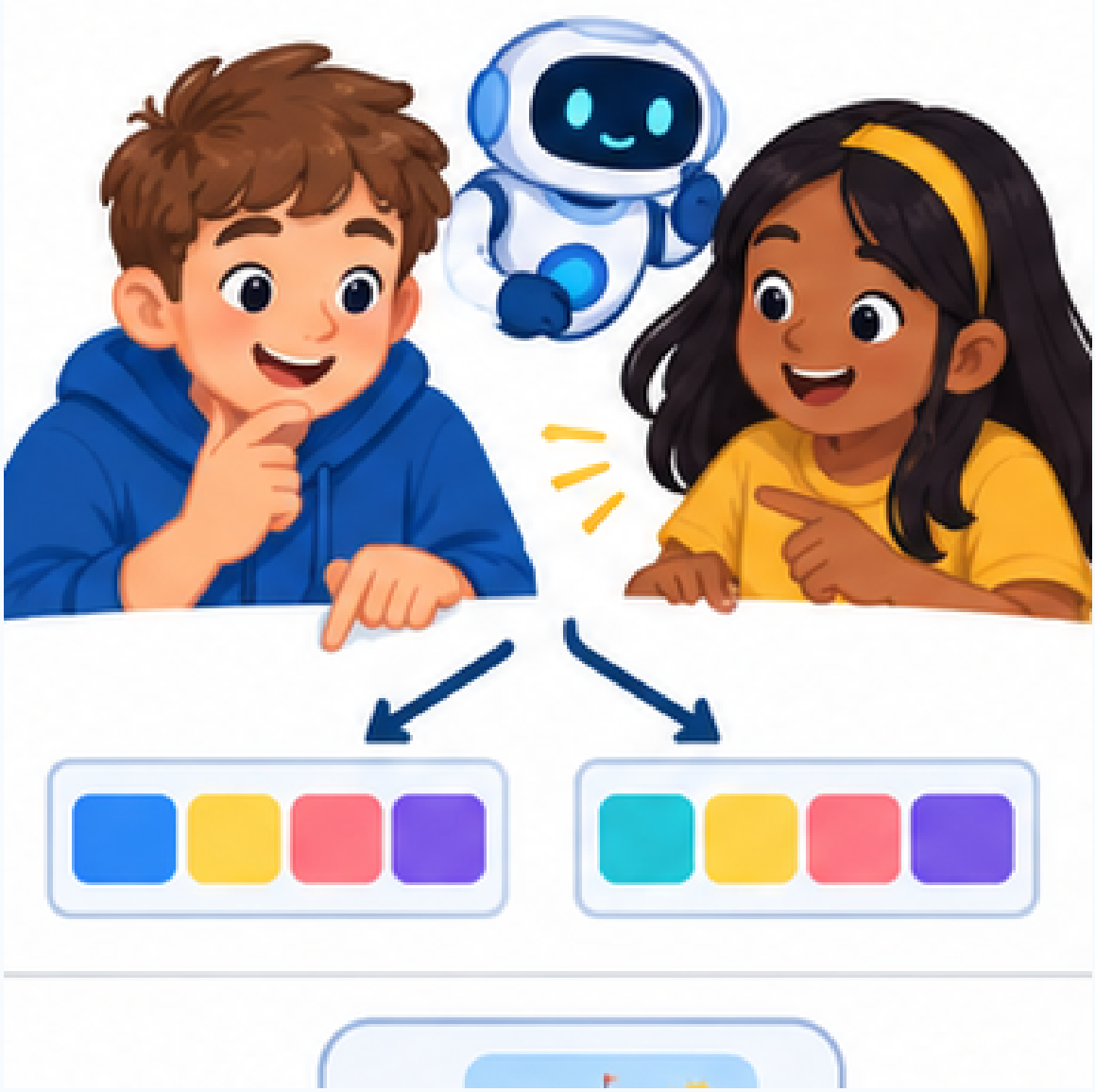
Notice the evidence

> Investigate the supplied Text as Pieces case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use text as pieces to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Tokens need not be words. A classroom **token**izer can use an exact, case-sensitive **vocabulary** and greedily choose the longest matching **token** from left to right. This is a toy rule, not a claim about every production **token**izer.

Define IDs: 0=end, 1=cat, 2=c, 3=a, 4=t, 5=s, 6=space, 7=sat. The space **token** represents one literal space. The end **token** is a special terminator, not ordinary text.

At each position choose the longest matching ordinary **token**. If no **token** matches, stop and report the unsupported character and its position. Do not delete the character or guess a replacement.

IDs identify entries; a larger ID does not mean a **token** is more valuable or more similar.

STUDENT LESSON · 3 OF 18

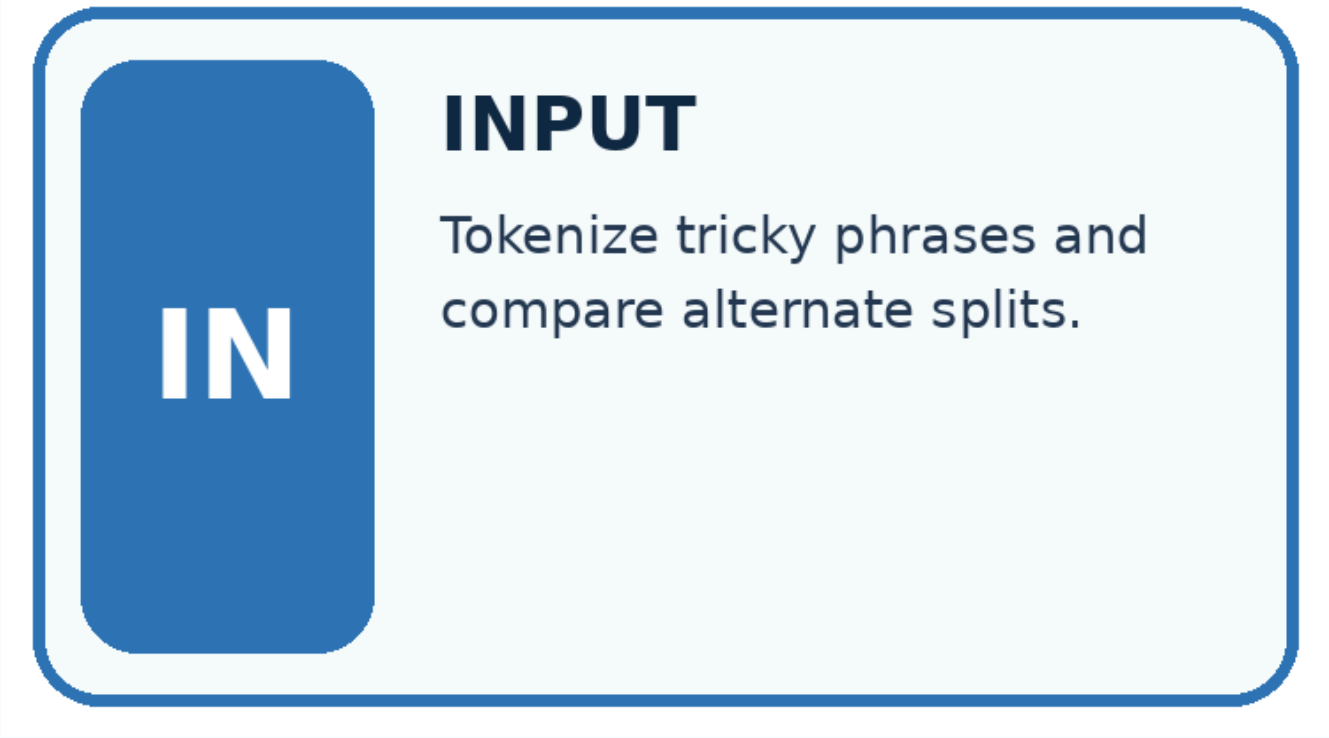
See how it works

> Follow the mechanism for Text as Pieces. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

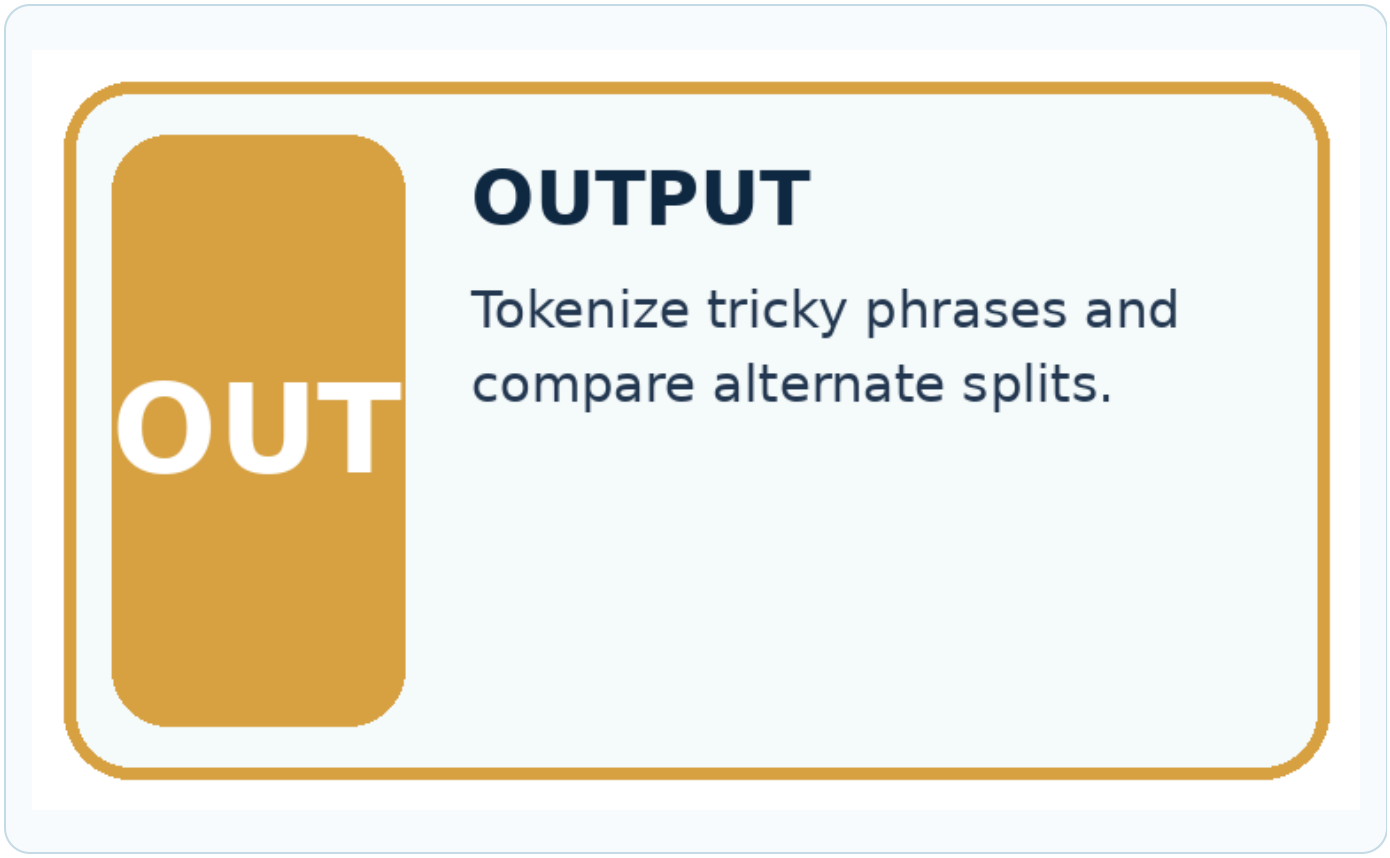
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
Language systems process tokens and patterns rather than whole human meanings.

↓


OUT
OUTPUT
Tokenize tricky phrases and compare alternate splits.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Remaining text	Chosen token	ID
cats sat	cat	1
s sat	s	5
sat	space	6
sat	sat	7

The **encoding** of cats sat is [1,5,6,7], followed by [0] only if the protocol appends an end **token**. **Decoding** concatenates the ordinary **token** strings to recover cats sat.

Count four text **token**s, or five sequence items with the end marker included; always state the counting convention. The two written words do not imply two **token**s.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Exact text: cats sat	Longest-match encoding	IDs 1, 5, 6, 7
IDs 1, 5, 6, 7	Concatenate token strings	Exact text: cats sat
Exact text: Cat	No match for uppercase C	Report unsupported input

Encoding preserves supported characters and makes unsupported input explicit.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

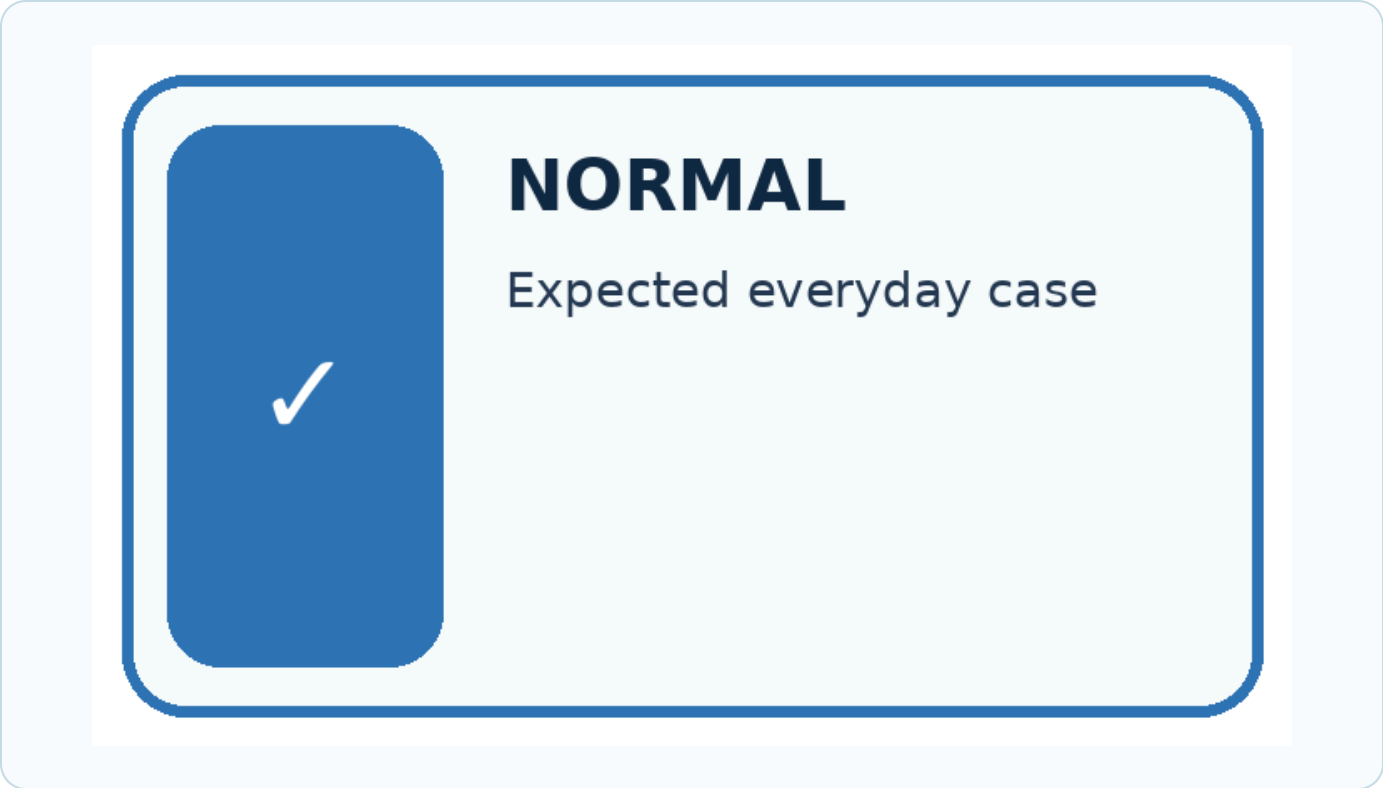
> Compare the supplied cases for Text as Pieces. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

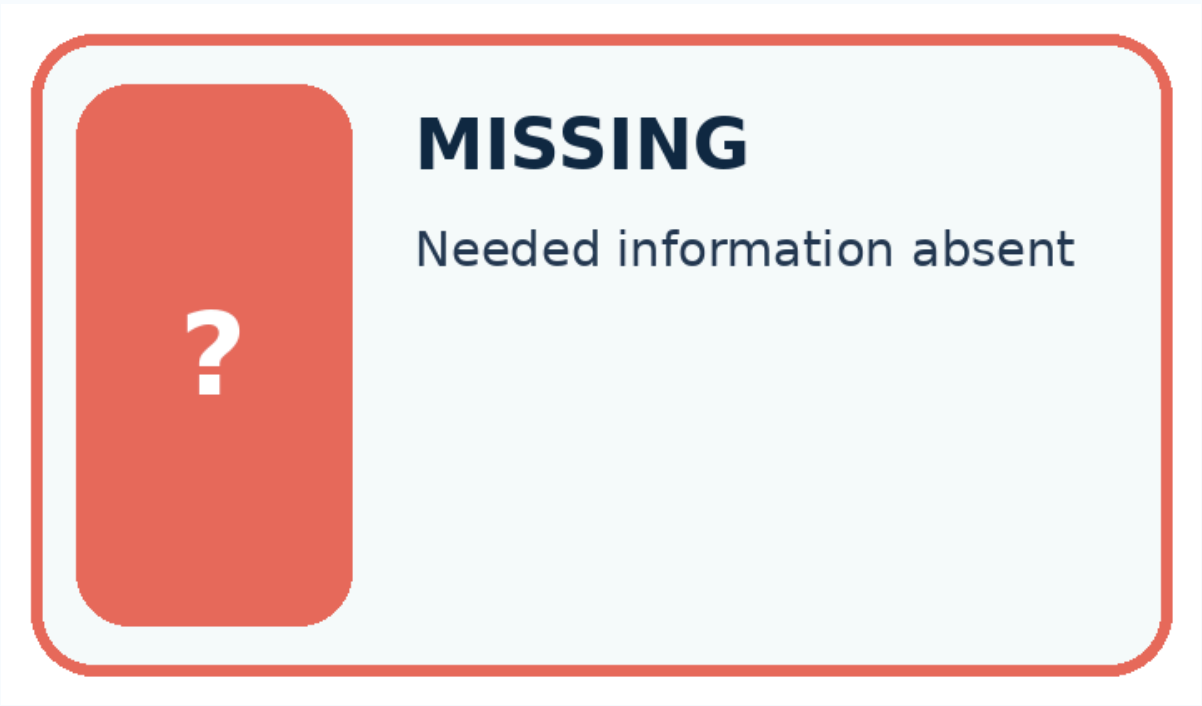
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Use a character **vocabulary** covering b,o,k,s and space instead.

Predicted result

Prediction: The same phrase uses ten character **token**s.

Why the result follows

Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Text as Pieces to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

TOKEN

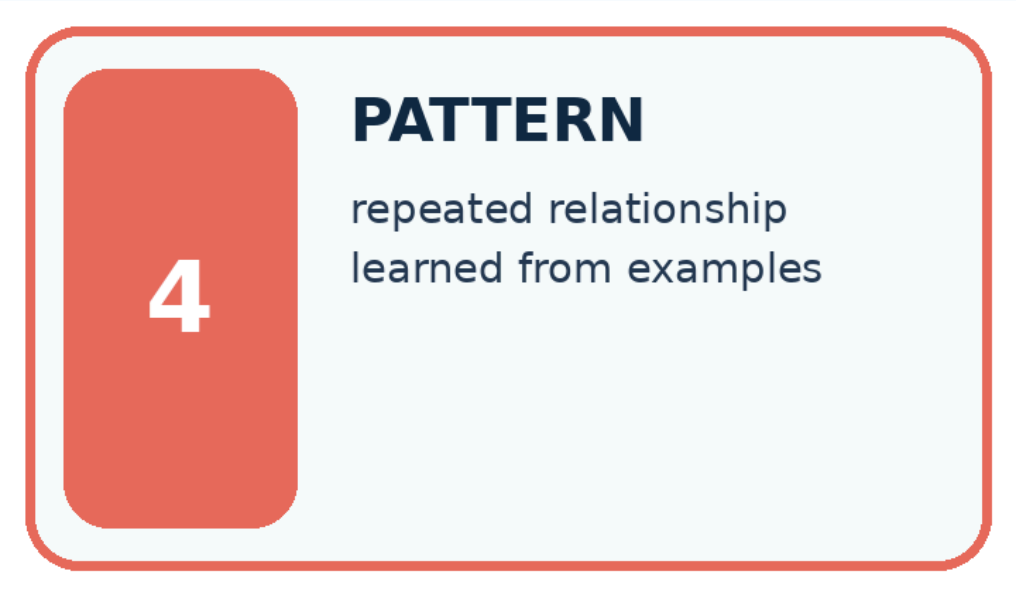
piece of text processed by a model

2

TOKENIZATION

rules for splitting text into pieces





Word	My own example	Not this / boundary
token	A unit of encoded text processed by a language model, which may be a word, part of a word or punctuation.	
tokenization	Converting text into the token units of a particular tokenizer.	
context	Information available around a task that helps interpret or answer it.	

Word	My own example	Not this / boundary
pattern	A relationship or regularity observed in examples.	

Words used in the closer explanation

Word	Meaning
Vocabulary	The permitted tokens and their identifiers
Encoding	Mapping source text into a token sequence under a specified rule
Decoding	Mapping a token sequence back to text under a specified rule

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Text as Pieces, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 25 • WORKED EVIDENCE

1. ‘unhelpful’ might split as ‘un’ + ‘help’ + ‘ful’.
2. A name or emoji may use unexpected token pieces.
3. Token boundaries need not match whole words.
4. Meaning emerges from patterns across context, not from one token acting like a dictionary.

1. ‘unhelpful’ might split as ‘un’ + ‘help’ + ‘ful’.

2. A name or emoji may use unexpected **token** pieces.

3. **Token** boundaries need not match whole words.

4. Meaning emerges from **pattern**s across **context**, not from one **token** acting like a dictionary.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Worked explanation · Text as Pieces

A character-only **vocabulary** c,a,t,s,space could encode cats sat as c,a,t,s,space,s,a,t: eight text **token**s. The original **vocabulary** uses four. Both decode to the same source text.

Sequence length is a property of the text, **vocabulary** and **encoding** rule together. Counts from different **token**izers are not automatically comparable.

In this **vocabulary**, sat is chosen over s because it is longer. The toy longest-match rule is defined explicitly. Real **token**izers may use different algorithms, byte **encoding**s and special-**token** policies.

Never infer an unknown **token**izer's count from word count alone.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Text as Pieces

A fictional **token**izer converts a short catalogue phrase into **token**s.

The facts we will use

Supplied facts

Vocabulary entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

Supplied facts

Text is “books book”. Exclamation marks have no **token**; zero-based character positions are used.

AI ACADEMY / AI-2/25

Text as Pieces

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read prefix

At position 0, longest matching entry is books, not book.



2. Encode phrase

Tokens: books, space-character, book; three tokens.



3. Decode

Joining token text returns exactly books book.



4. Unknown character

For books book!, the ! at position 10 has no token: stop with error.

CHANGE ONE THING

Use a character vocabulary covering b,o,k,s and space instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read prefix	At position 0, longest matching entry is books, not book.
2. Encode phrase	Tokens: books, space-character, book; three tokens.
3. Decode	Joining token text returns exactly books book.
4. Unknown character	For books book!, the ! at position 10 has no token: stop with error.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

Preserve spaces during **decoding** so the reconstructed text matches the original exactly.

What this example does not establish

Keep the stated limit

This deterministic toy **token**izer omits real subword **training**, Unicode handling and model-specific special **token**s.

Keep the stated limit

Read the labelled picture: Why is the unknown ! at position 10 rather than position 9?

Keep the stated limit

Change one thing: Use a character **vocabulary** covering b,o,k,s and space instead. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Text as Pieces claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Every token is exactly one English word. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Decoding round trips check representation, not factual meaning. A supported spelling can be false; an unsupported character can be important.

Record the **vocabulary** version, rule and special-**token** convention when comparing counts or reproducing model **input**s. Whitespace, case and punctuation are part of this exact toy **input** contract.

Exit: a program drops ! from cat! and announces a successful round trip. Identify the changed information and the honest response.

STUDENT LESSON · 8 OF 18

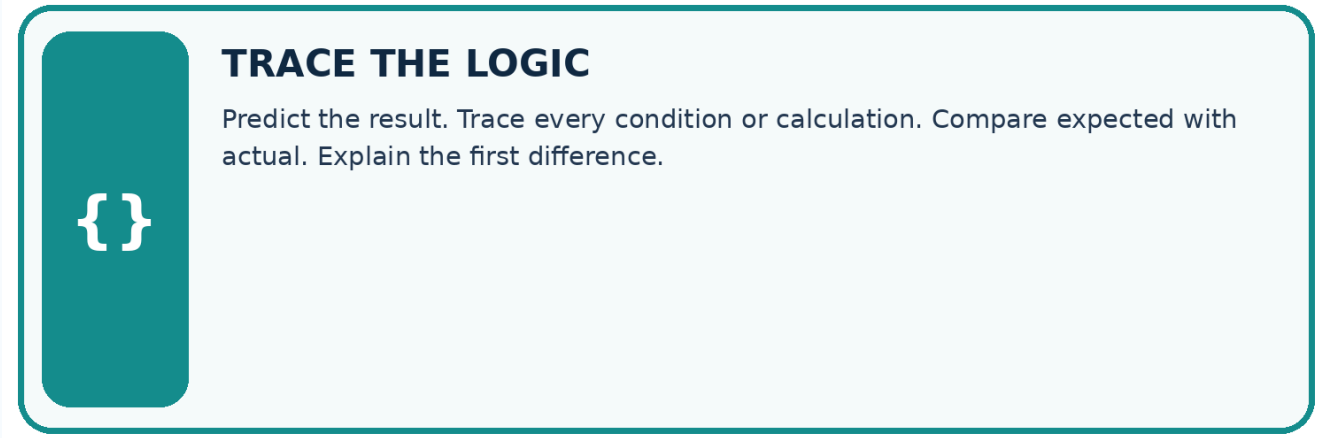
Practise together

> Practise Text as Pieces with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Supported case: cat encodes [1]; c at encodes [2,6,3,4]. A space changes both the character sequence and the possible matches. The **input** Cat is unsupported at position 0 because uppercase C has no entry.

The **input** cat! encodes cat, then fails on ! at position 3; a partial **encoding** must not be reported as complete success.

Guided: encode c at, count its text tokens and explain how cat! should be reported.

Practise with less help: Encode and decode

Use workbook W2 for the connected example "**Token**isation and **vocabulary**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Token**isation and **vocabulary**. Proposed explanation: "A **token** always equals a word". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why is the unknown ! at position 10 rather than position 9?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

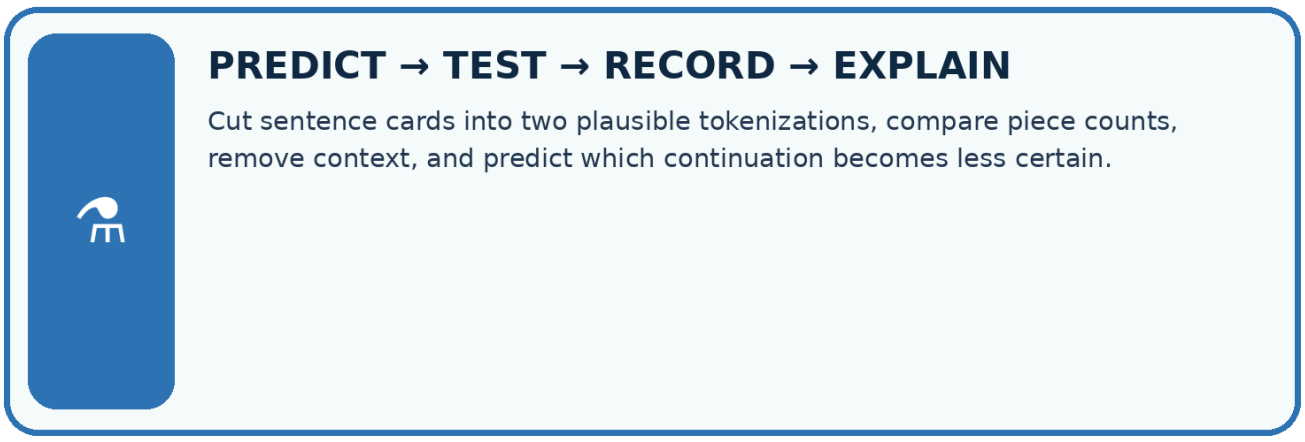
Run the investigation

> Investigate Text as Pieces. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Cut sentence cards into two plausible **tokenization**s, compare piece counts, remove **context**, and predict which continuation becomes less certain.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

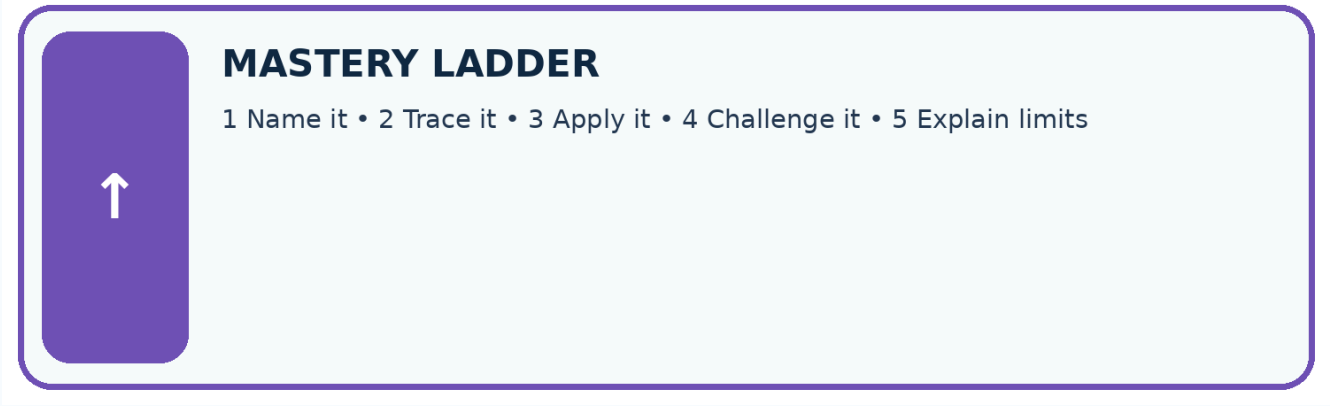
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Text as Pieces. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

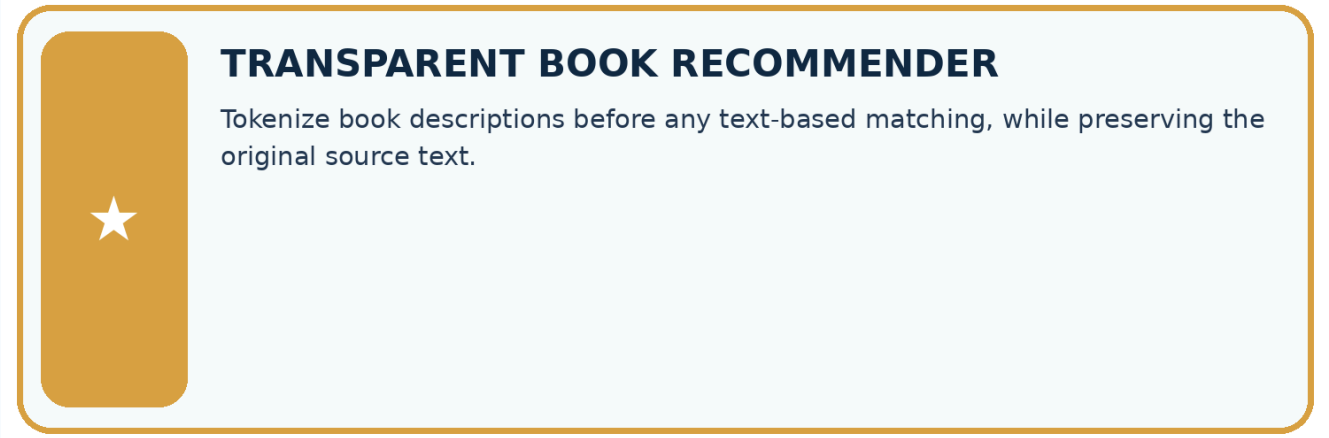
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT **Token**ize book descriptions before any text-based matching, while preserving the original source text.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Document the recommender's text representation and unknown-**token** behavior rather than assuming every model counts words identically.

STUDENT LESSON · 12 OF 18

Show what you know

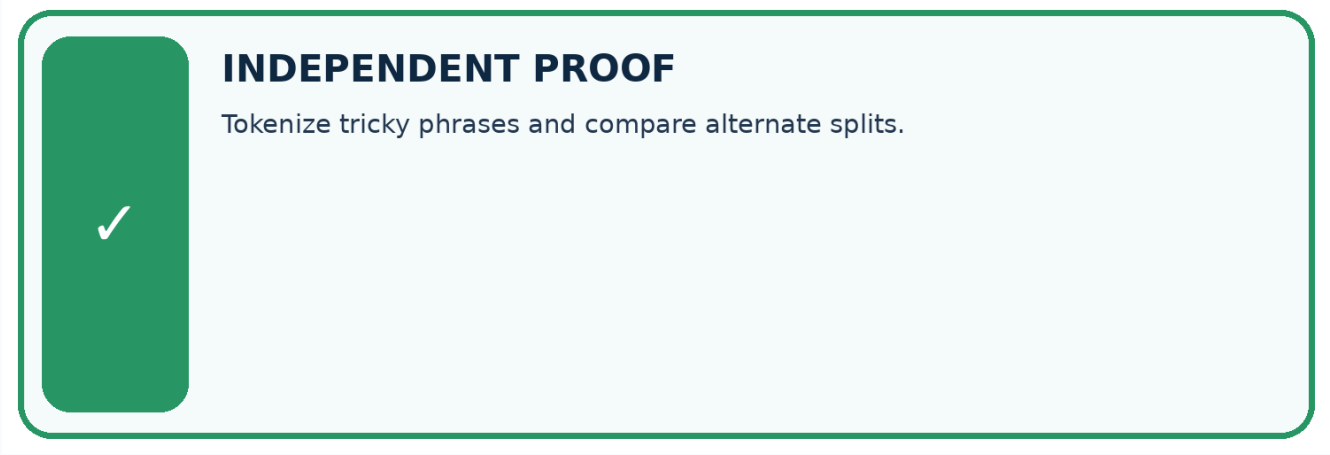
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE **Token**ize tricky phrases and compare alternate splits.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Text as Pieces case for supported practice. Repeated examples do not become fresh assessments.

Week 25 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Toy **token**izer A splits rainbows as [rain][bow][s]; B uses [rainbows].

The question

Count **token**s and state the limit of the example.

Worked explanation

Three versus one. **Token** count depends on **token**izer; **token**s need not equal words or letters.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

One word can have different token counts

Evidence or stage	Value or result
Tokenizer A	rain / bow / s
Tokenizer B	rainbows

Toy **token**izer A splits rainbows as [rain][bow][s]; B uses [rainbows].

Count **token**s and state the limit of the example.

Three versus one. **Token** count depends on **token**izer; **token**s need not equal words or letters.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Trace a new input independently

Under the original **vocabulary** encode sat cats, decode it and count text **token**s. Then trace sat cat! and state the exact failure position using zero-based character positions.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 25 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A classroom **token**izer splits sunflower into sun and flower. How many pieces result? Must every real **token**izer split it the same way?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Tokenize book descriptions before any text-based matching, while preserving the original source text.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Toy **token**izer A splits 'unhelpful' into un/help/ful; B keeps it whole. Count **token**s for each and explain why a three-word sentence need not use three **token**s.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Text as Pieces, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional **token**izer converts a short catalogue phrase into **token**s.

By the end, I can	How I will show it
Encode and decode text under an explicit toy vocabulary	Explain the deciding step, show a trace or test, and state one limit.
Explain how segmentation changes sequence length	Explain the deciding step, show a trace or test, and state one limit.
Handle unsupported characters without silently deleting information	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Text as Pieces without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Encode and decode text under an explicit toy **vocabulary**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional **token**izer converts a short catalogue phrase into **token**s.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? For books book!, the ! at position 10 has no **token**: stop with error.

> 1. Unknown character

> 2. Decode

> 3. Encode phrase

> 4. Read prefix

> **My selected conclusion, deciding fact and explanation**

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Unknown character For books book!, the ! at position 10 has no **token**: stop with error.

A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

Preserve spaces during **decoding** so the reconstructed text matches the original exactly.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Use a character **vocabulary** covering b,o,k,s and space instead. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The same phrase uses ten character **token**s.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The same phrase uses ten character **token**s. There are five letters, one space and four letters; its decoded text remains unchanged.

Repair a missing building block

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why do k=1 and k=3 disagree for the original query?. Return to this week and explain what you changed.

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can **80%** response be reported as **80%** of the whole grade responding?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

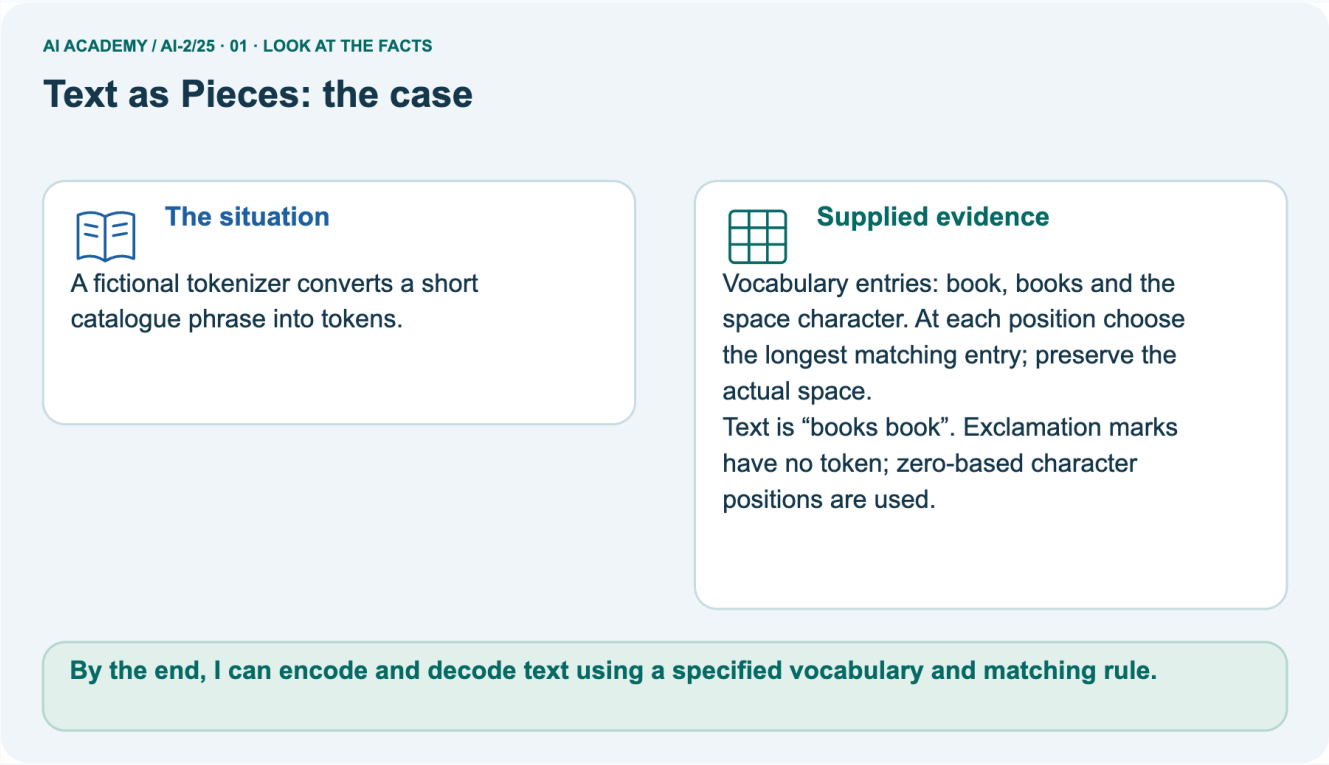
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Text as Pieces graphic using its labels, deciding evidence and relevant limitation.

01. Text as Pieces: the case



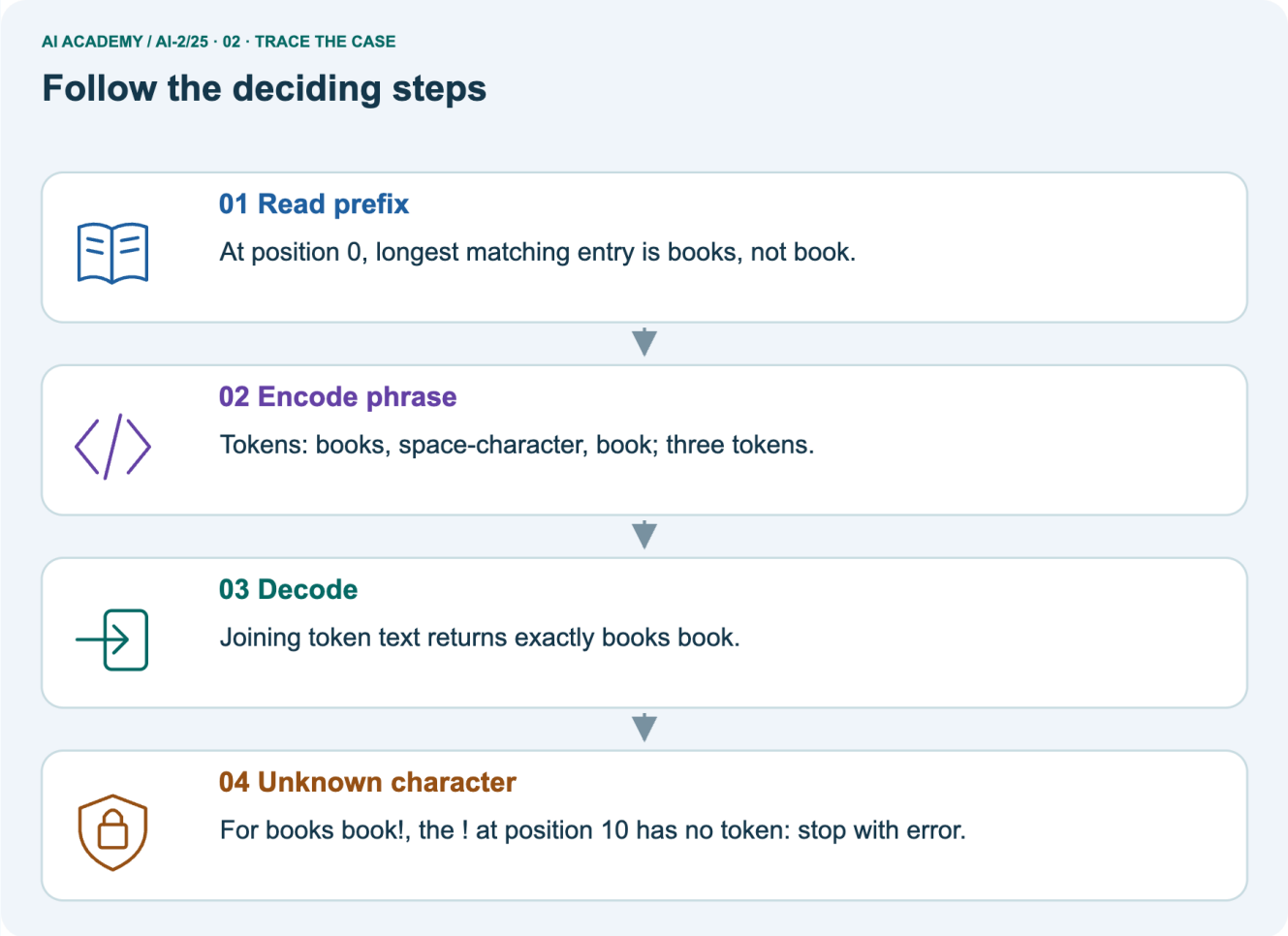
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional **token**izer converts a short catalogue phrase into **token**s. Supplied facts: **Vocabulary** entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

Text is “books book”. Exclamation marks have no **token**; zero-based character positions are used.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

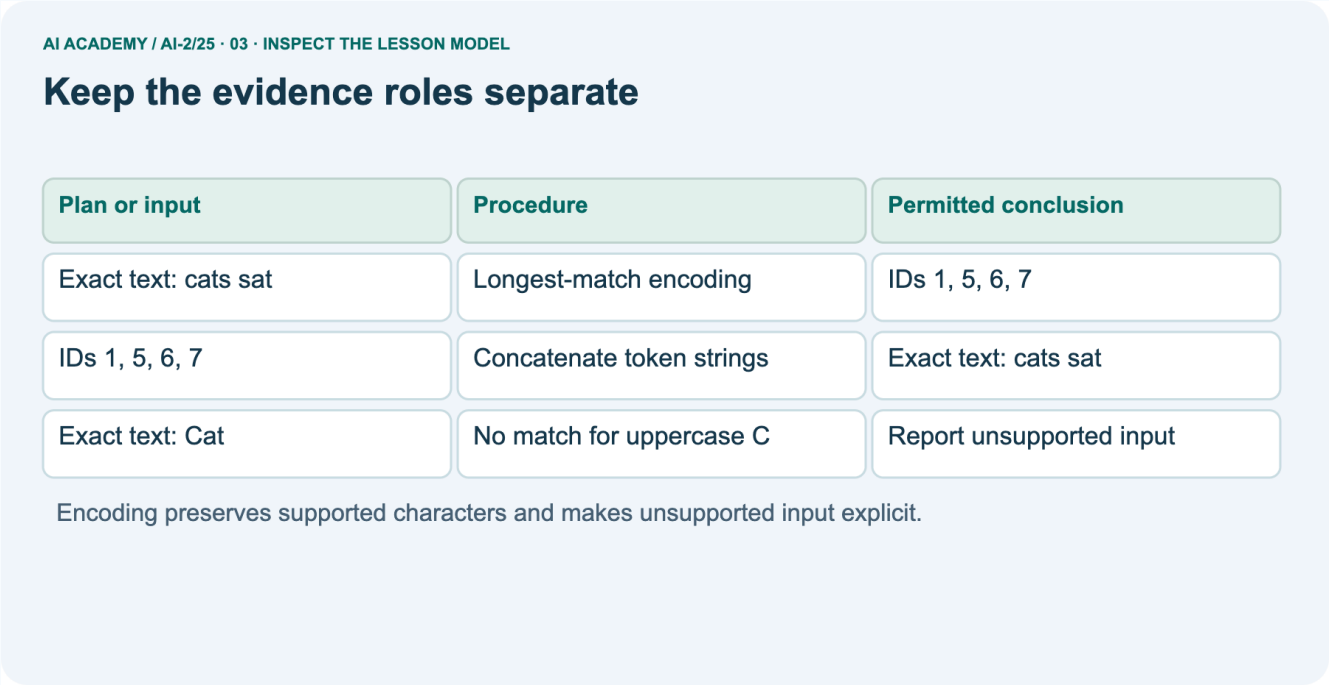


Follow the labelled picture with the matching explanation. Open the vector diagram

Read prefix: At position 0, longest matching entry is books, not book. → Encode phrase: **Token**s: books, space-character, book; three **token**s. → Decode: Joining **token** text returns exactly books book. → Unknown character: For books book!, the ! at position 10 has no **token**: stop with error.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep the evidence roles separate



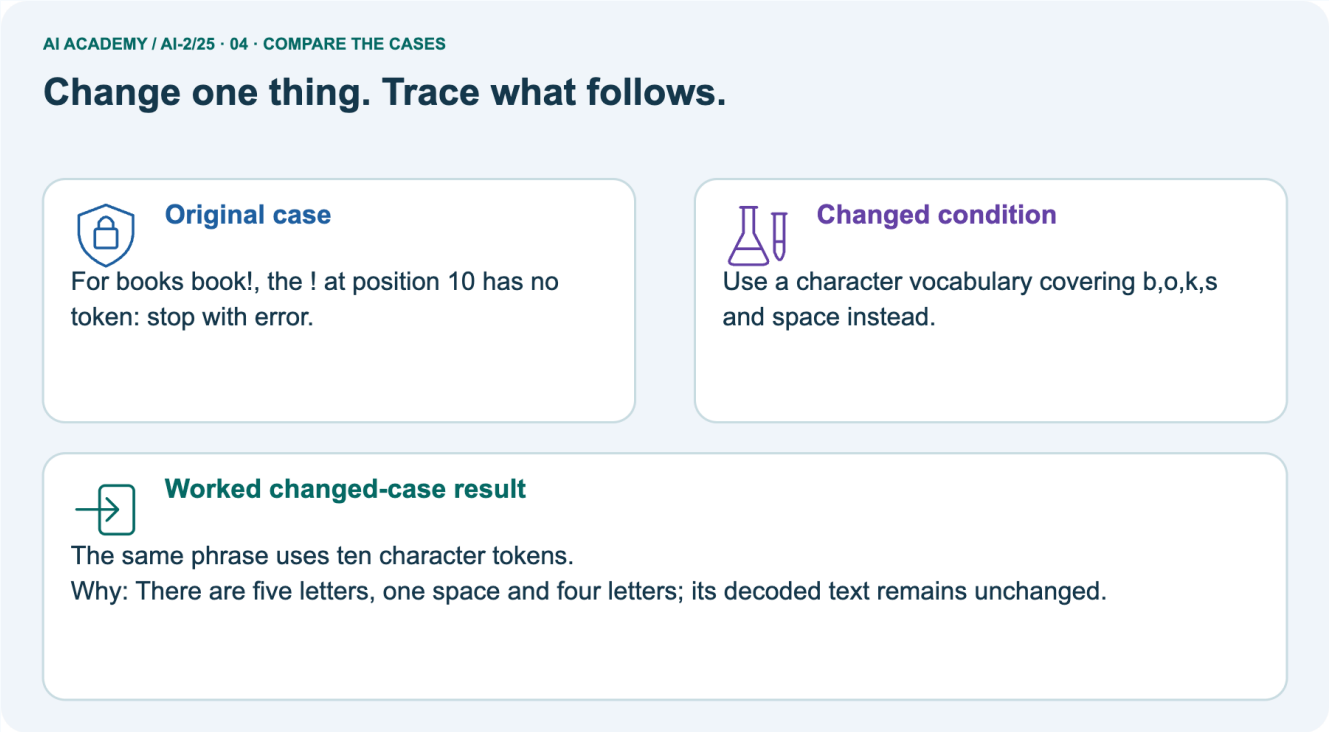
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. Exact text: cats sat | Longest-match **encoding** | IDs 1, 5, 6, 7. IDs 1, 5, 6, 7 | Concatenate **token** strings | Exact text: cats sat.

Exact text: Cat | No match for uppercase C | Report unsupported **input**

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



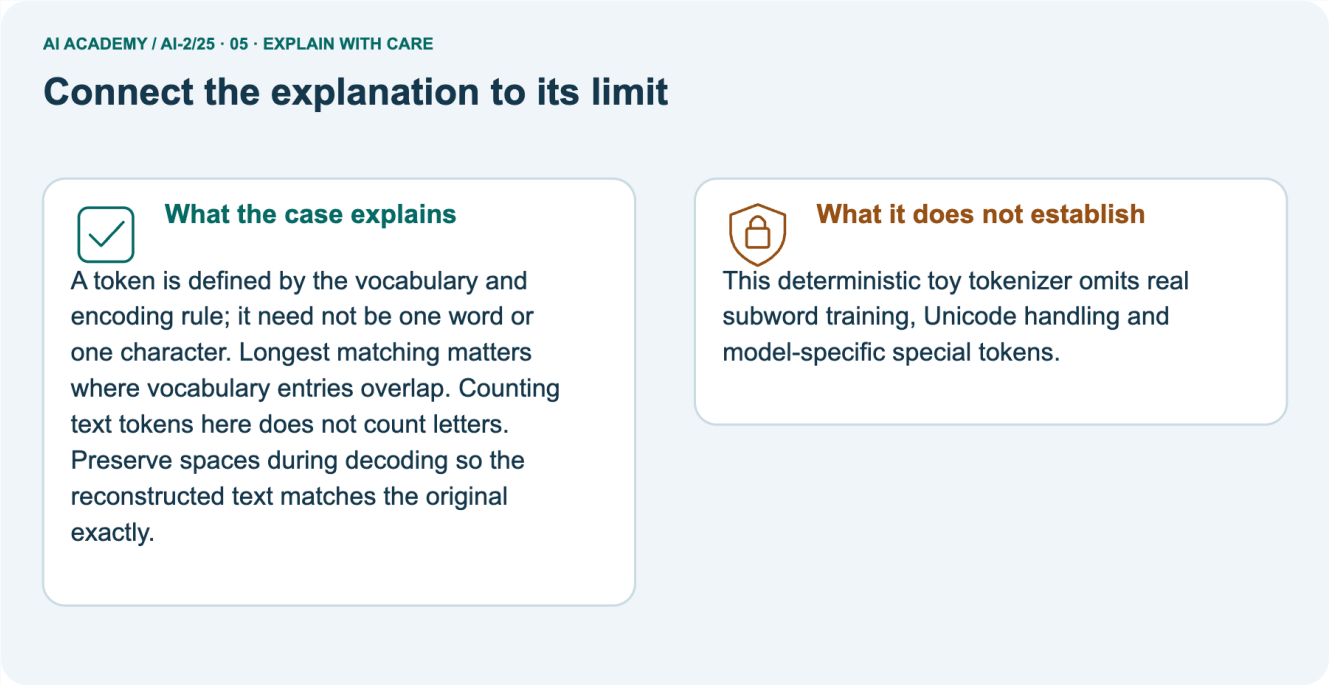
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: For books book!, the ! at position 10 has no **token**: stop with error. Changed condition: Use a character **vocabulary** covering b,o,k,s and space instead. Expected result: The same phrase uses ten character **token**s.

Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



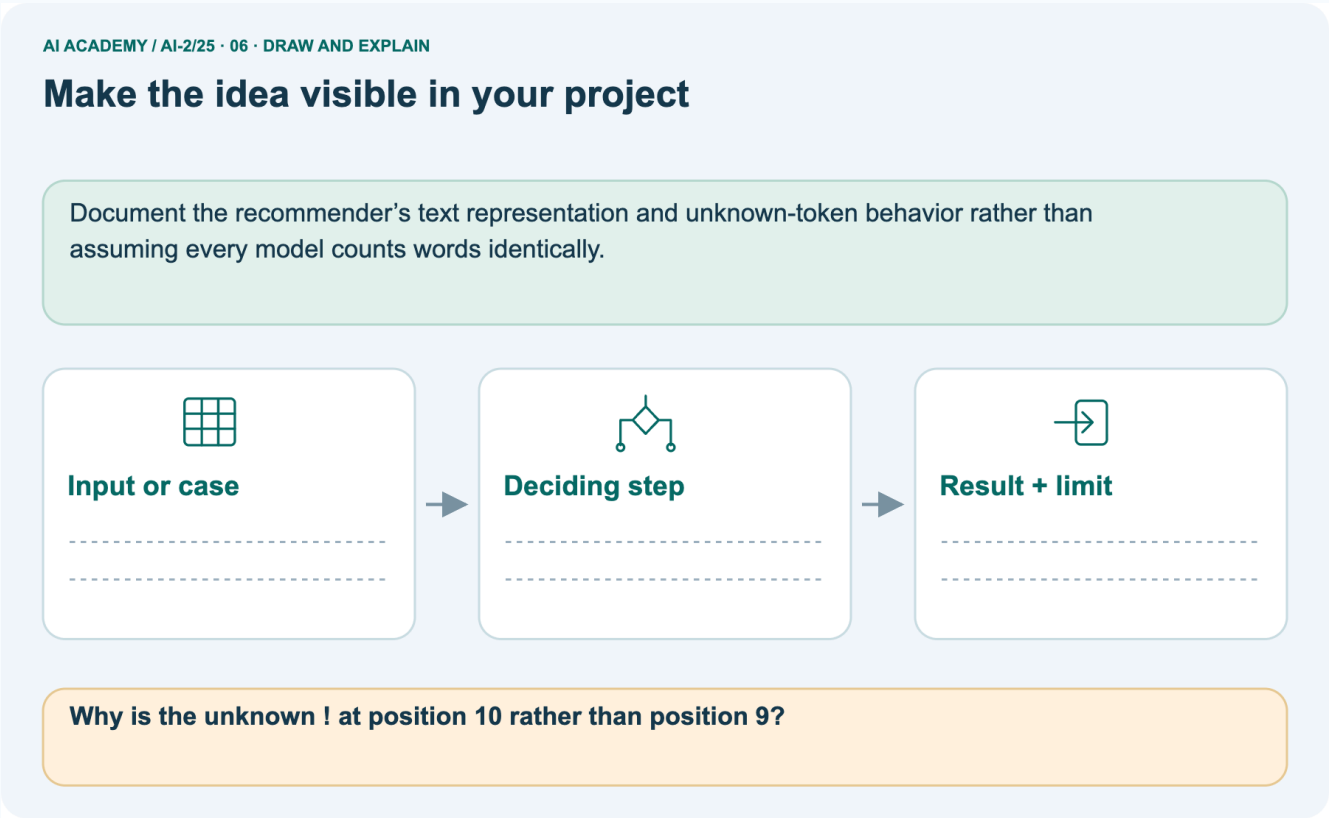
Follow the labelled picture with the matching explanation. Open the vector diagram

A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

Preserve spaces during **decoding** so the reconstructed text matches the original exactly. Limit: This deterministic toy **token**izer omits real subword **training**, Unicode handling and model-specific special **token**s.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document the recommender's text representation and unknown-**token** behavior rather than assuming every model counts words identically. Draw your **input**, deciding step, result and limit.

Supported question: Why is the unknown ! at position 10 rather than position 9?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.